

Syntax -Parsing I

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Week 5: Lecture 2

Grammar Rewrite Rules

$S \rightarrow NP VP$

$S \rightarrow Aux NP VP$

$S \rightarrow VP$

$NP \rightarrow Det NOM$

$NOM \rightarrow Noun$

$NOM \rightarrow Noun NOM$

$VP \rightarrow Verb$

$VP \rightarrow Verb NP$

$Det \rightarrow that \mid this \mid a \mid the$

$Noun \rightarrow book \mid flight \mid meal \mid man$

$Verb \rightarrow book \mid include \mid read$

$Aux \rightarrow does$

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$S \rightarrow NP VP$

$\rightarrow Det NOM VP$

$\rightarrow The NOM VP$

$\rightarrow The Noun VP$

$\rightarrow The man VP$

$\rightarrow The man Verb NP$

$\rightarrow The man read NP$

$\rightarrow The man read Det NOM$

$\rightarrow The man read this NOM$

$\rightarrow The man read this Noun$

$\rightarrow The man read this book$

S $\rightarrow$ NP VP
 $\rightarrow$ Det NOM VP
 $\rightarrow$ *The* NOM VP
 $\rightarrow$ *The* Noun VP
 $\rightarrow$ *The man* VP
 $\rightarrow$ *The man* Verb NP
 $\rightarrow$ *The man read* NP
 $\rightarrow$ *The man read* Det NOM
 $\rightarrow$ *The man read this* NOM
 $\rightarrow$ *The man read this* Noun
 $\rightarrow$ *The man read this book*

Parse Tree

S $\rightarrow$ NP VP

$\rightarrow$ Det NOM VP

$\rightarrow$ *The* NOM VP

$\rightarrow$ *The* Noun VP

$\rightarrow$ *The man* VP

$\rightarrow$ *The man* Verb NP

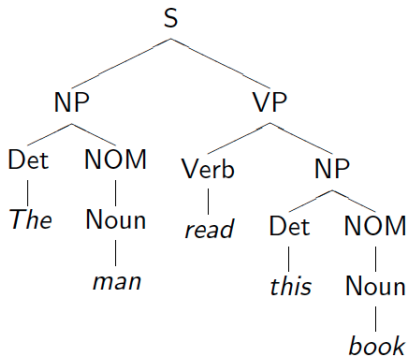
$\rightarrow$ *The man read* NP

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What are the constraints? “book that flight”

- There must be three leaves, *book*, *that* and *flight*
- The tree must have one root, the start symbol S
- Give rise to two search strategies: *top-down* (goal-oriented) and *bottom-up* (data-directed)

Grammar

S → NP VP

S → Aux NP VP

S → VP

NP → Pronoun

NP → Proper-Noun

NP → Det Nominal

Nominal → Noun

Nominal → Nominal Noun

Nominal → Nominal PP

VP → Verb

VP → Verb NP

VP → VP PP

PP → Prep NP

Lexicon

Det → the | a | that | this

Noun → book | flight | meal | money

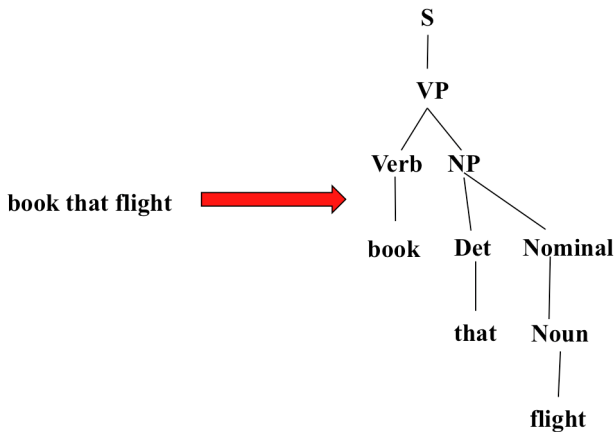
Verb → book | include | prefer

Pronoun → I | he | she | me

Proper-Noun → Houston | NWA

Aux → does

Prep → from | to | on | near | through



Top-Down Parsing

- Searches for a parse tree by trying to build upon the root node S down to the leaves

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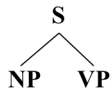
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- Trees are grown downward until they eventually reach the POS categories at the bottom
- Trees whose leaves fail to match the words in the input can be rejected

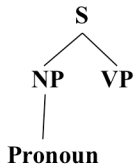
Top-Down Parsing

S

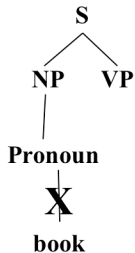
Top-Down Parsing



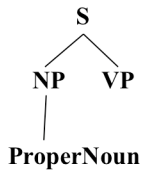
Top-Down Parsing



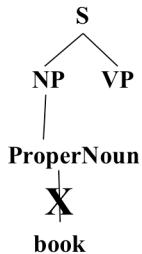
Top-Down Parsing



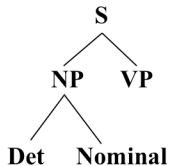
Top-Down Parsing



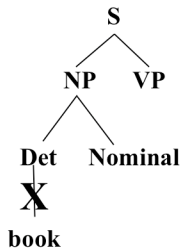
Top-Down Parsing



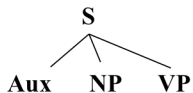
Top-Down Parsing



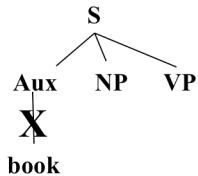
Top-Down Parsing



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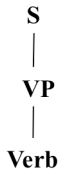
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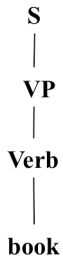
Top-Down Parsing

S
|
VP

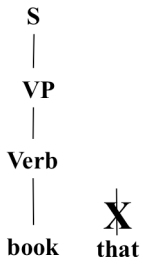
Top-Down Parsing



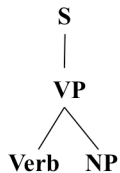
Top-Down Parsing



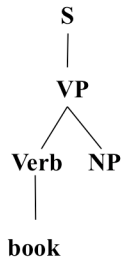
Top-Down Parsing



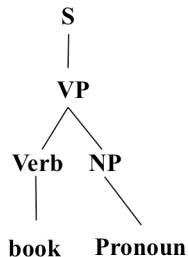
Top-Down Parsing



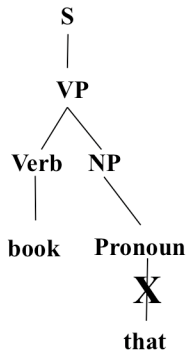
Top-Down Parsing



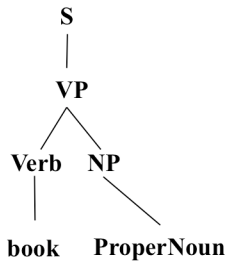
Top-Down Parsing



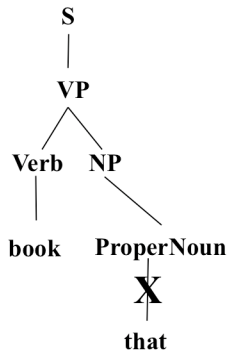
Top-Down Parsing



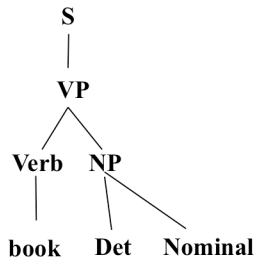
Top-Down Parsing



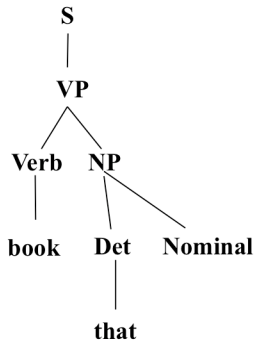
Top-Down Parsing



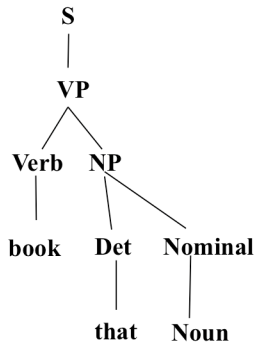
Top-Down Parsing



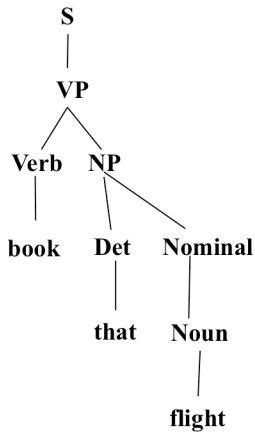
Top-Down Parsing



Top-Down Parsing



Top-Down Parsing



- The parser starts with the words of the input, and tries to build trees from the words up, by applying rules from the grammar one at a time
- Parser looks for the places in the parse-in-progress where the right-hand-side of some rule might fit.

Bottom-Up Parsing

book that flight

Bottom-Up Parsing

Noun

|

book

that

flight

Bottom-Up Parsing

Nominal

|

Noun

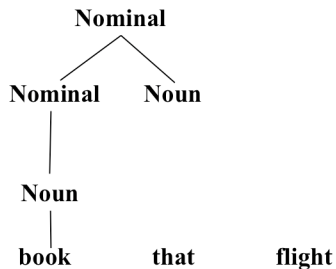
|

book

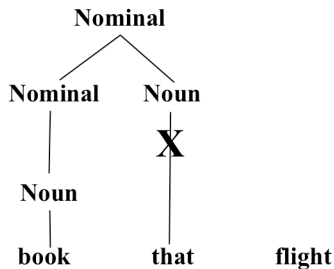
that

flight

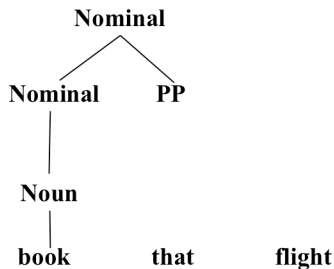
Bottom-Up Parsing



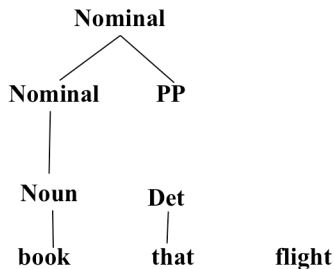
Bottom-Up Parsing

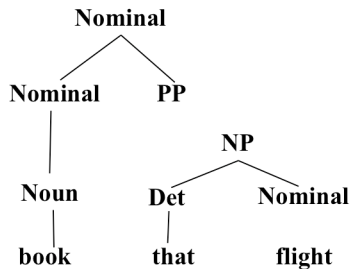


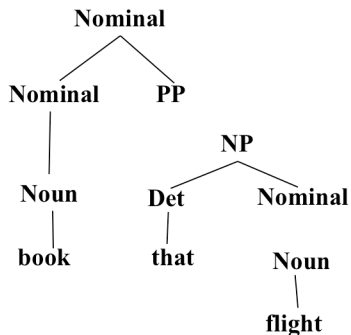
Bottom-Up Parsing

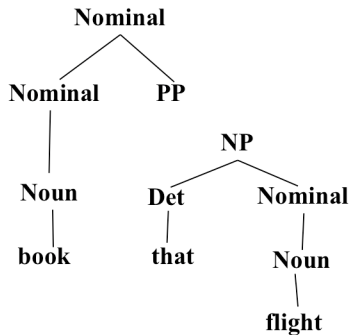


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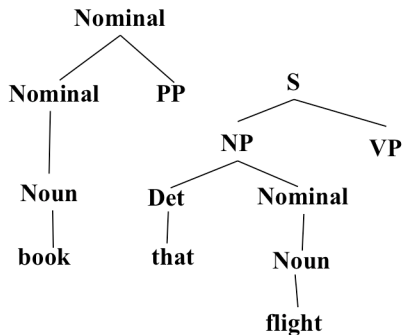




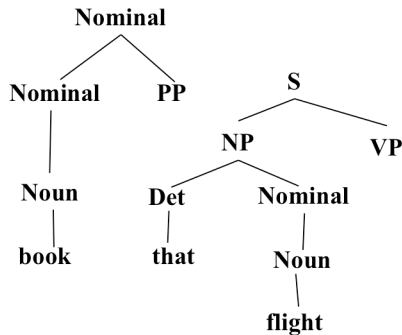




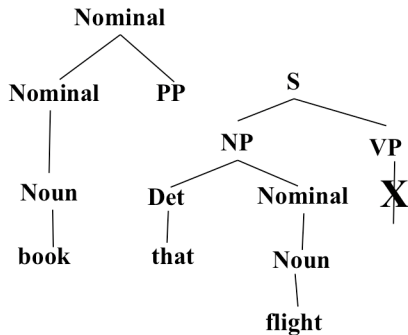
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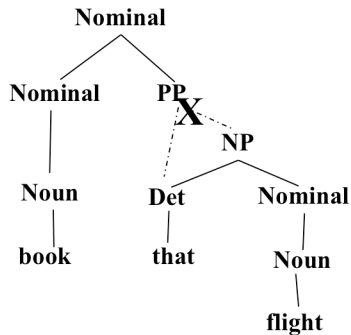
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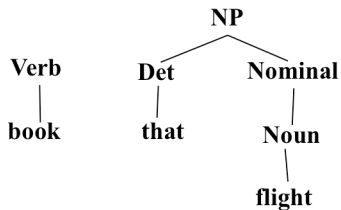
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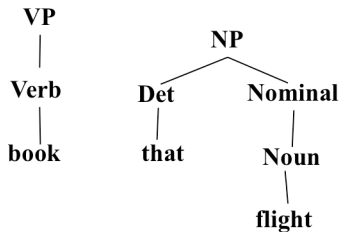
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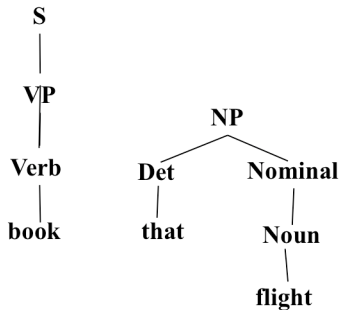
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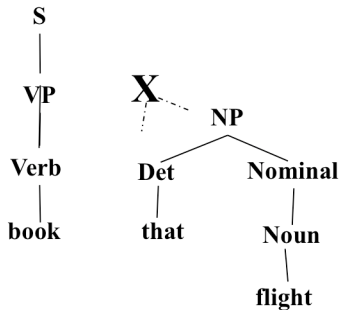
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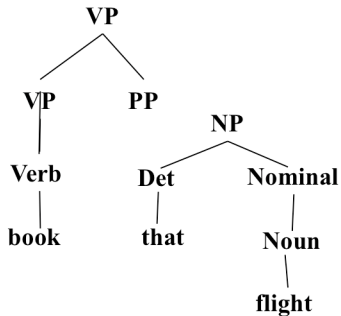
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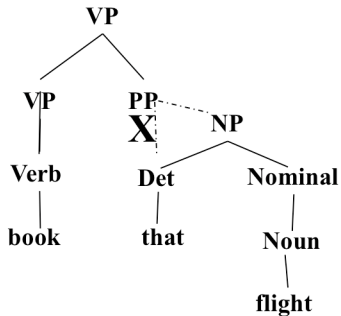
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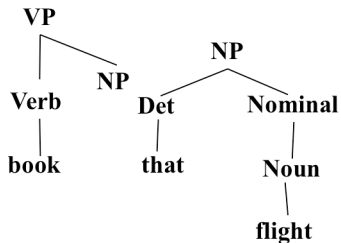
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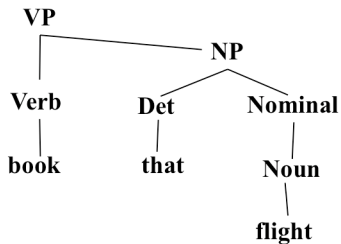


Bottom-Up Parsing

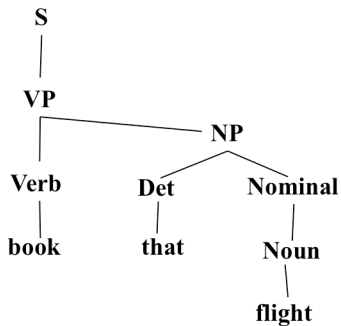


Bottom-Up Parsing





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Top-Down vs. Bottom-Up

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- Top down never explores options that will not lead to a full parse, but can explore many options that never connect to the actual sentence.
- Bottom up never explores options that do not connect to the actual sentence but can explore options that can never lead to a full parse.
- Relative amounts of wasted search depend on how much the grammar branches in each direction.

Dynamic Programming Parsing

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- Caching (memoizing) critical to obtaining a polynomial time parsing (recognition) algorithm for CFGs.
- Dynamic programming algorithms based on both top-down and bottom-up search can achieve $O(n^3)$ recognition time where n is the length of the input string.

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- More generally, *chart parsers* retain completed phrases in a chart and can combine top-down and bottom-up searches.

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- Parse bottom-up storing phrases formed from all substrings in a triangular table (chart)

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Chomsky Normal Form

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